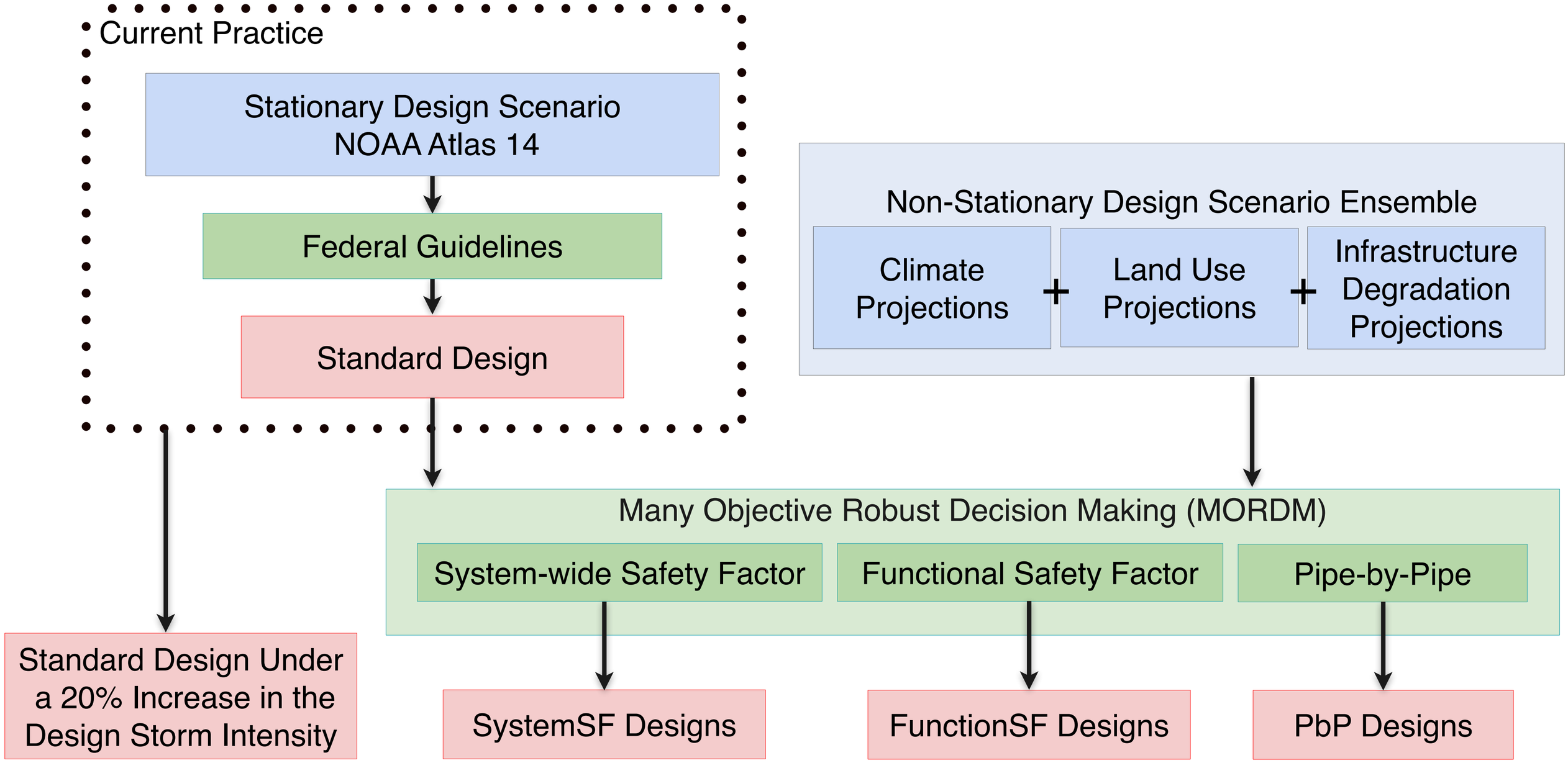
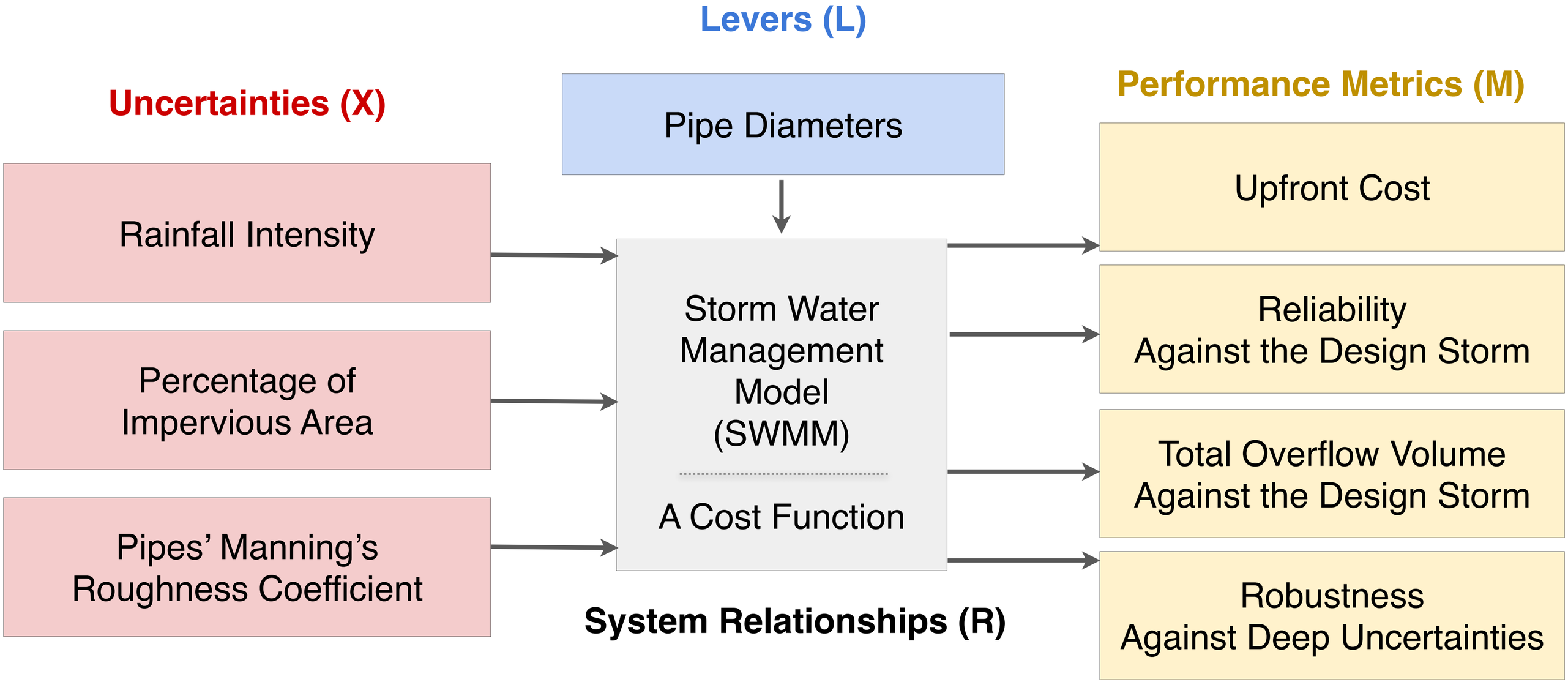
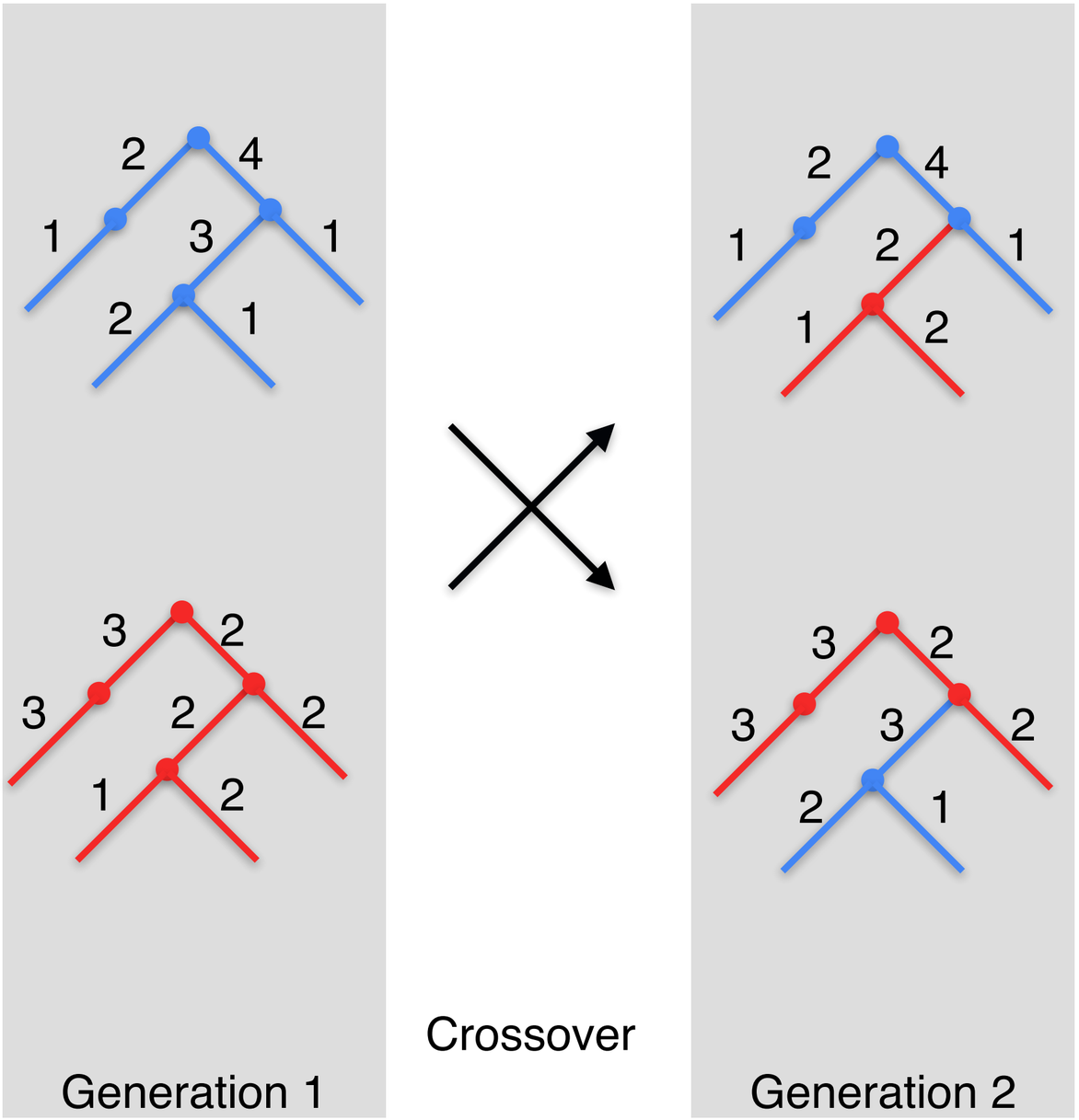
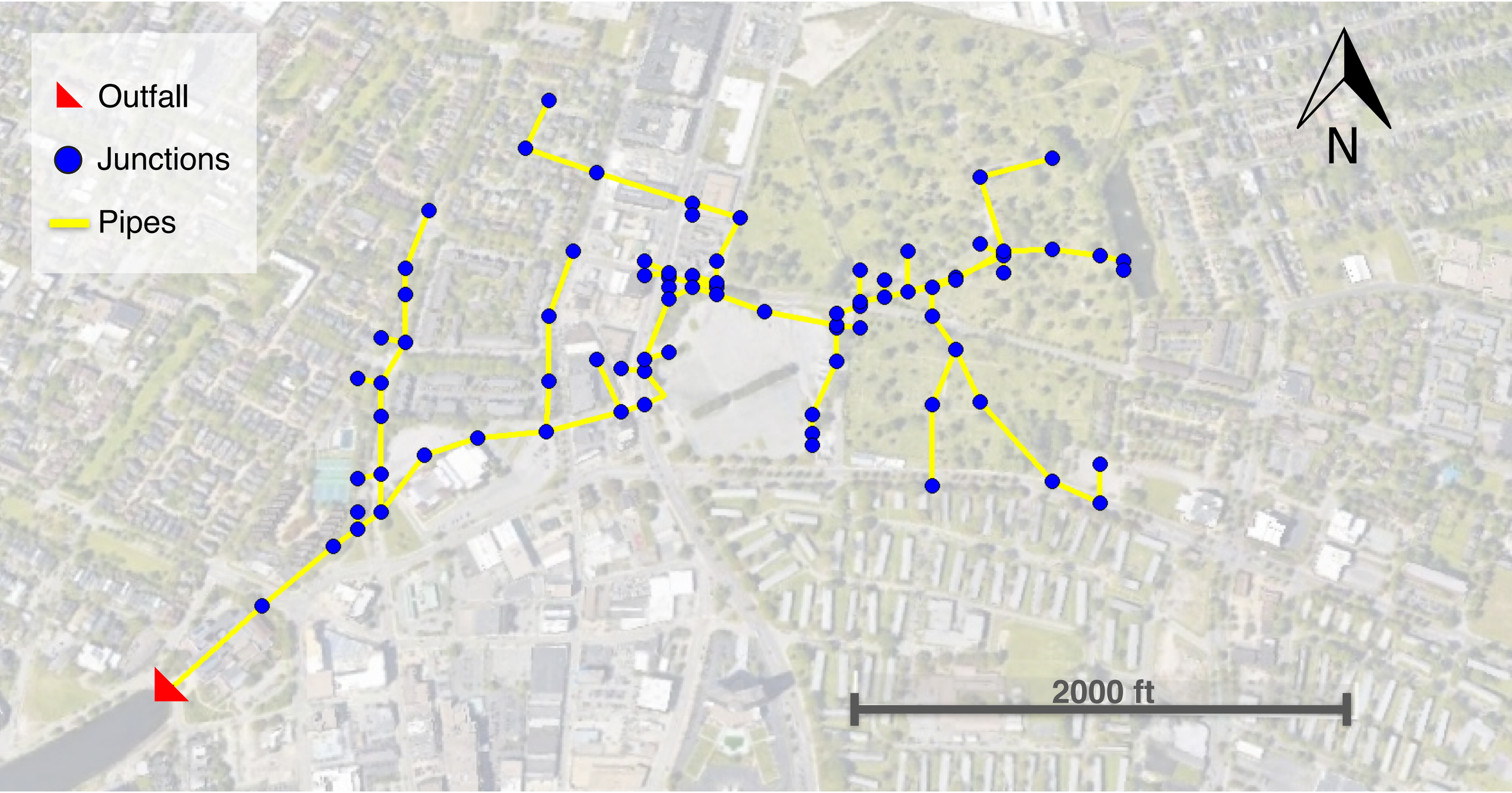
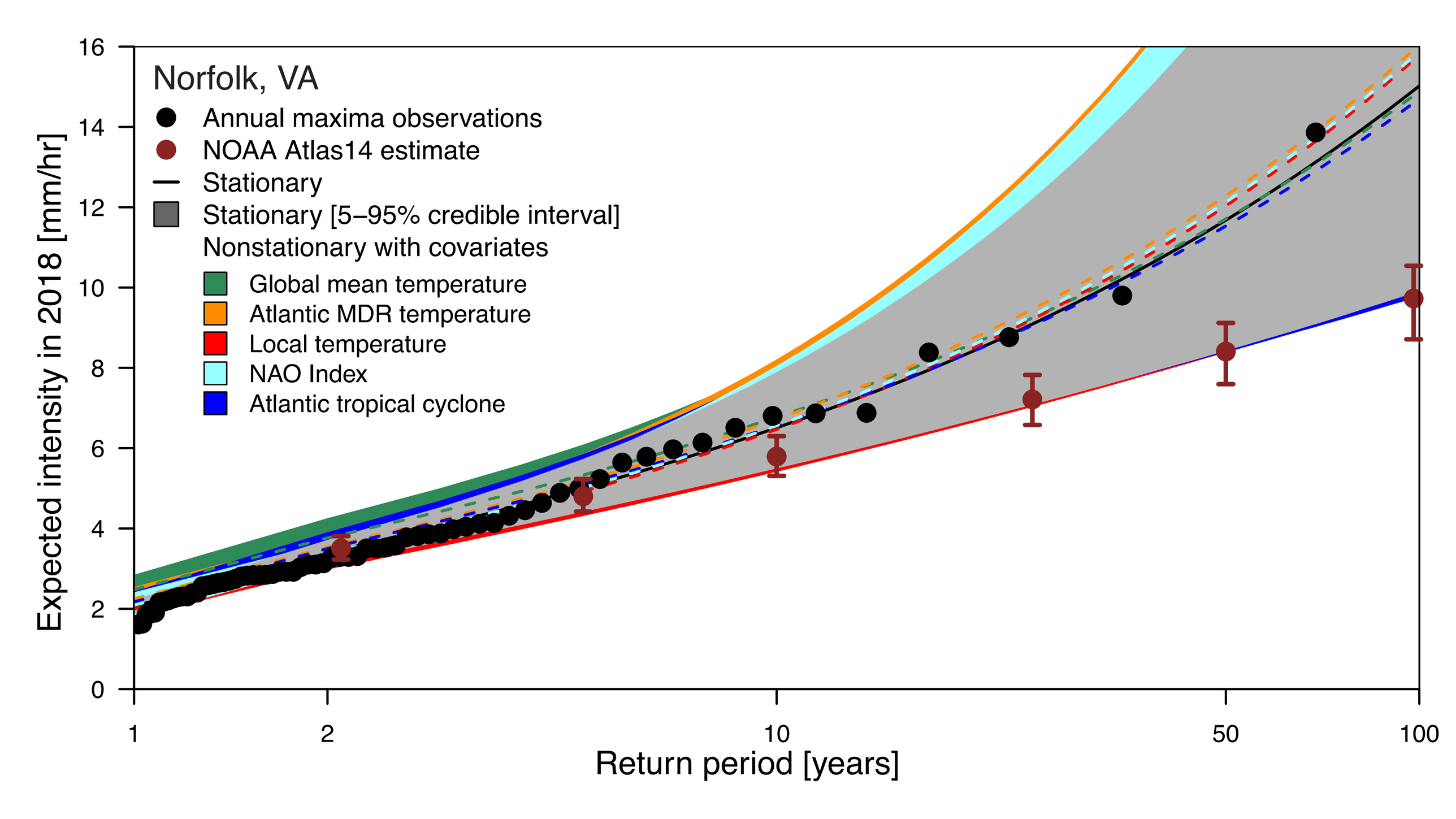

Current Practice
Stationary Design Scenario
NOAA Atlas 14
Federal Guidelines
Standard Design
Non-Stationary Design Scenario Ensemble
Climate Projections
+
Land Use Projections
+
Infrastructure Degradation Projections
Many Objective Robust Decision Making (MORDM)
System-wide Safety Factor
Functional Safety Factor
Pipe-by-Pipe
Standard Design Under a 20% Increase in the Design Storm Intensity
SystemSF Designs
FunctionSF Designs
PbP Designs
Input Data
Design Strategies
Design Outcomes
Data Flow


Levers (L)
Uncertainties (X)
Performance Metrics (M)
Pipe Diameters
Rainfall Intensity
Percentage of Impervious Area
Pipes' Manning's Roughness Coefficient
Storm Water Management Model (SWMM)
A Cost Function
Upfront Cost
Reliability Against the Design Storm
Total Overflow Volume Against the Design Storm
Robustness Against Deep Uncertainties
System Relationships (R)


2
4
1
3
1
2
1
3
2
3
2
2
1
2
Crossover
2
4
1
2
1
1
2
3
2
3
3
2
2
1
Generation 1
Generation 2


Outfall
Junctions
Pipes
N
2000 ft


Norfolk, VA
Annual maxima observations
NOAA Atlas14 estimate
Stationary
Stationary [5–95% credible interval]
Nonstationary with covariates
Global mean temperature
Atlantic MDR temperature
Local temperature
NAO Index
Atlantic tropical cyclone
Expected intensity in 2018 [mm/hr]
0
2
4
6
8
10
12
14
16
1
2
10
50
100
Return period [years]